ciliated tentacles that function in feeding (see Figure 32.12a). Individuals in other phyla, including molluscs and annelids, go through a distinctive developmental stage called the **trochophore larva** (see Figure 32.12b)—hence the name lophotrochozoan.

The clade name **Ecdysozoa** refers to a characteristic shared by nematodes, arthropods, and some of the other ecdysozoan phyla that are not included in our survey. These animals secrete external skeletons (exoskeletons); the stiff covering of a cricket and the flexible cuticle of a nematode are examples. As the animal grows, it molts, squirming out of its old exoskeleton and secreting a larger one. The process of shedding the old exoskeleton is called *ecdysis*. Though named for this characteristic, the clade was proposed mainly on the basis of molecular data that support the common ancestry of its members. Furthermore, some taxa excluded from this clade by their molecular data, such as certain species of leeches, do in fact molt.

Future Directions in Animal Systematics

While many scientists think that current evidence supports the phylogeny in Figure 32.11, aspects continue to be debated. Although it can be frustrating that the phylogenies in textbooks are not set-in-stone truths, the uncertainty inherent in these diagrams is a healthy reminder that science is an ongoing, dynamic process of inquiry. We'll conclude with two questions that are the focus of ongoing research.

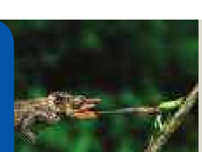
1. **Are ctenophores basal metazoans?** Many researchers have concluded that sponges are basal metazoans (see Figure 32.11). This conclusion was supported in a 2017 phylogenomic analysis, but several other studies have placed the comb jellies (phylum Ctenophora) at the base of the animal tree. In addition to the most recent phylogenomic results, data consistent with placing sponges at the base of the animal tree include fossil steroid evidence, molecular clock analyses, the morphological similarity of sponge collar cells to the cells of choanoflagellates (see Figure 32.3), and the fact that sponges are one of the few animal groups that lack tissues (as might be expected for basal animals). Ctenophores, on the other hand, have tissues, and their cells do not resemble the cells of choanoflagellates. At present, the idea that ctenophores are basal metazoans remains an intriguing but controversial hypothesis.
2. **Are the Acoela basal bilaterians?** A series of recent papers—including a 2016 phylogenomic study—have indicated that flatworms in phylum Acoela are basal bilaterians, as shown in Figure 32.11. A different conclusion was supported by one recent analysis, which placed members of the Acoela within Deuterostomia. Researchers are sequencing the genomes of additional species in the phylum and in closely related groups to provide a more definitive test of the hypothesis that the Acoela are basal bilaterians. If further evidence supports this hypothesis, bilaterians may have descended from a common ancestor that resembled this group of living flatworms—that is, an ancestor that had a simple nervous system, a saclike gut with a single opening (the "mouth"), and no excretory system.

CONCEPT CHECK 32.4

1. Describe the evidence that cnidarians share a more recent common ancestor with other animals than with sponges.
2. **WHAT IF?** Suppose ctenophores are basal metazoans and sponges are the sister group of all remaining animals. Under this hypothesis, redraw Figure 32.11 and discuss whether animals with tissues would form a clade.
3. **MAKE CONNECTIONS** Based on the phylogeny in Figure 32.11 and the information in Figure 25.11, evaluate this statement: "The Cambrian explosion actually consists of three explosions, not one."

For suggested answers, see Appendix A.

32 Chapter Review



Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

SUMMARY OF KEY CONCEPTS

To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to goo.gl/zkjz9t.

CONCEPT 32.1

Animals are multicellular, heterotrophic eukaryotes with tissues that develop from embryonic layers

(pp. 674–675)

- Animals are heterotrophs that ingest their food.
- Animals are multicellular eukaryotes. Their cells are supported and connected to one another by collagen and other structural proteins located outside the cell membrane. Nervous tissue and muscle tissue are key animal features.
- In most animals, **gastrulation** follows the formation of the **blastula** and leads to the formation of embryonic tissue layers. Most animals have *Hox* genes that regulate the development of body form. Although *Hox* genes have been highly conserved over the course of evolution, they can produce a wide diversity of animal morphology.

? *Describe key ways that animals differ from plants and fungi.*